Part 2 of 2

Next-Generation SOLAR POWER TECHNOLOGY

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Quantum-dot solar cells based on perovskites materials will convert not just rooftops but complete buildings, including windows, into solar energy generators. Up to 26% efficiency has been achieved by mechanically combining perovskite with silicon solar cells

Quantum dots are already being used in quantum-dot televisions. These have revolutionised the display industry with ultra-high-definition colours and greatly increased effective viewing angles, eliminating the need for rare earth elements—minerals where China has a virtual monopoly in the market.

Quantum dots, however, can also be used to absorb light to boost the output of photovoltaics, photocatalysts, light sensors and other optoelectronic devices. In fact, quantum-dot solar cells can absorb energy 24 × 7. While conventional silicon modules can absorb visible light during peak times of the day, quantum-dot solar cells can absorb ultraviolet through visible to infrared lighting spectrum to produce power day and night.

Through a process called multiple exciton generation, quantum dots capture excess photon energy that is normally lost to heat generation. The incident light radiations enter through the transparent electrode of a quantum-dot solar cell onto a light-absorbing layer of dots and create electron-hole pairs (e^-/h^+). The charged particles then separate and eventually travel to their respective electrodes, thereby producing electric current.

Quantum dots are made from electro-optically active materials, whose size as well as composition controls the photon energy that they interact with, allowing their absorption (and emission) wavelength to be tuned by particle size. These can be tuned to deliver multiple electron-hole pairs from one photon—a process called multiple exciton generation that has allowed quantum-dot solar cells to operate at 10 per cent efficiency.

In quantum-dot solar cells, the function of light-absorbing material is carried out by nanocrystals of semiconducting metal chalcogenides such as CdS, CdSe, PbS and PbSe. Like other emerging photovoltaic technologies, quantum-dot cells can be prepared via low-cost solution-phase chemistry methods and are amenable to high-speed printing techniques.

The structure of quantum dots is peculiar with generally cadmium selenide as the inner core and a $Cd_{1-x}Zn_xS$ outer shell, coated in silica to avoid oxidation. The outer shell acts like an absorber. When a photon interacts with a quantum dot, the electron transfers from its valence band into the conduction band, leaving a hole in the valence band. The cell is designed such that the shell only absorbs high-energy photons, and the new photon from the core can propagate via internal reflections throughout the glass and quantum-dot layers. Finally, the propagating photons would reach the glass edges, where one or more solar cells could pick them up.

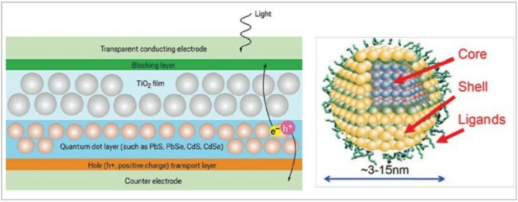

Fig. 7: Structure and working mechanism of a quantum-dot solar cell